(v) $\cos\frac{3A}{2}\cos\frac{5A}{2} = \frac{1}{2}\cos\left(\frac{3A}{2}+\frac{5A}{2}\right) + \cos\left(\frac{3A}{2}-\frac{5A}{2}\right)$

$= \frac{1}{2}\left[\cos\frac{8A}{2}+\cos\left(\frac{-2A}{2}\right)\right]$

$= \frac{1}{2}[\cos 4A + \cos(-A)]$

$= \frac{1}{2}[\cos 4A + \cos A]$

(vi) $\cos 9\theta \sin 6\theta = \frac{1}{2}[\sin(9\theta+6\theta) - \sin(9\theta-6\theta)]$

$= \frac{1}{2}[\sin(15\theta) - \sin 3\theta]$

(vii) $2\cos 13A \sin 15A = \sin(13A+15A) - \sin(13A-15A)$

$= \sin 28A + \sin 2A$

Example 4.15

Show that $\sin 20^\circ \sin 40^\circ \sin 80^\circ = \frac{\sqrt{3}}{8}$

Solution

LHS $= \sin 20^\circ \sin 40^\circ \sin 80^\circ$

$= \sin 20^\circ \sin(60^\circ - 20^\circ)\sin(60^\circ + 20^\circ)$

$= \sin 20^\circ[\sin^2 60^\circ - \sin^2 20^\circ]$

$= \sin 20^\circ\left[\frac{3}{4} - \sin^2 20^\circ\right]$

$= \sin 20^\circ\left[\frac{3 - 4\sin^2 20^\circ}{4}\right]$

$= \frac{3\sin 20^\circ - 4\sin^3 20^\circ}{4} = \frac{\sin 3(20^\circ)}{4}$

$= \frac{\sin 60^\circ}{4} = \frac{\sqrt{3}/2}{4} = \frac{\sqrt{3}}{8}$.

Example 4.16

Show that $\sin 20^\circ \sin 40^\circ \sin 60^\circ \sin 80^\circ = \frac{3}{16}$

Solution

L.H.S $= \sin 20^\circ \sin 40^\circ \sin 60^\circ \sin 80^\circ$

$= \sin 60^\circ \sin 20^\circ \sin(60^\circ - 20^\circ)\sin(60^\circ + 20^\circ)$

$= \frac{\sqrt{3}}{2}\sin 20^\circ(\sin^2 60^\circ - \sin^2 20^\circ)$

$= \frac{\sqrt{3}}{2}\sin 20^\circ\left(\frac{3}{4} - \sin^2 20^\circ\right)$

$= \left(\frac{\sqrt{3}}{2}\right)\left(\frac{1}{4}\right)(3\sin 20^\circ - 4\sin^3 20^\circ)$

$= \left(\frac{\sqrt{3}}{2}\right)\left(\frac{1}{4}\right)\sin 60^\circ = \left(\frac{\sqrt{3}}{2}\right)\left(\frac{1}{4}\right)\left(\frac{\sqrt{3}}{2}\right) = \frac{3}{16}$

= R.H.S.

Example 4.17

Prove that $\cos^2 A + \cos^2(A+120^\circ) + \cos^2(A-120^\circ) = \frac{3}{2}$

Solution

$\cos^2 A + \cos^2(A+120^\circ) + \cos^2(A-120^\circ)$

$= \cos^2 A + \cos^2(A+120^\circ) + (1 - \sin^2(A-120^\circ))$

(Since $\cos^2 A = 1 - \sin^2 A$)

$= \cos^2 A + 1 + \cos^2(A+120^\circ) - \sin^2(A-120^\circ)$

$= \cos^2 A + 1 + \cos[(A+120^\circ) + (A-120^\circ)]\cos[(A+120^\circ) - (A-120^\circ)]$

(Since $\cos^2 A - \sin^2 B = \cos(A+B)\cos(A-B)$)

$= \cos^2 A + 1 + \cos 2A \cos 240^\circ$

$= \cos^2 A + 1 + (2\cos^2 A - 1)\cos(180^\circ + 60^\circ)$

(since $2\cos^2 A = 1 + \cos 2A$)

$= \cos^2 A + 1 + (2\cos^2 A - 1)(-\cos 60^\circ)$

$= \cos^2 A + 1 + (2\cos^2 A - 1)\left(-\frac{1}{2}\right)$

$= \cos^2 A + 1 - \cos^2 A + \frac{1}{2} = \frac{3}{2}$

Example 4.18

Convert the following into the product of trigonometric functions.

(i) $\sin 9A + \sin 7A$ (ii) $\sin 7\theta - \sin 4\theta$

(iii) $\cos 8A + \cos 12A$ (iv) $\cos 4\alpha - \cos 8\alpha$

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